

Team Project 1 & Tutorial Preparation Master Guide: Linear Mathematical Models

Concordia University · Department of Building, Civil and Environmental Engineering
Instructor: Dr. A. Haghighat M. · Curriculum: Lecture 6 & Zill Section 2.7 · Deliverable: 1-Hour In-Tutorial Team Project 1

1. Team Project 1 — Operational Rules & Strategic Playbook

1.1 Official Academic Regulations

TEAM PROJECT 1 TUTORIAL SPECIFICATIONS	
Delivery Mode	In-person during scheduled tutorial session
Time Duration	Exactly 1 hour (60 minutes)
Team Size	Strictly 2 or 3 students (no solo, no 4+)
Submission Format	Physical paper submission at end of hour
Required Header	Full Names + Student IDs of ALL team members
Grading Policy	Identical grade assigned to all team members

1.2 The 60-Minute Team Execution Protocol

To ensure top marks within the 1-hour time constraint, divide responsibilities dynamically:

THE 60-MINUTE TUTORIAL TIME BUDGET		
Window	Phase	Team Deliverables
00–05 m	Read & Classify	Identify archetype (Growth, Cool, Tank, Circuit). List given numeric parameters.
05–15 m	Governing ODE & Initial Cond.	Write rate balance ($R_{in} - R_{out}$, KVL, etc) Define variables with units [lb, gal, s]
15–35 m	Analytic Solution	Calculate integrating factor $\mu(t)$ or separate variables. Integrate completely.
35–45 m	Evaluate Constants	Apply IVP conditions to isolate c and k .
45–55 m	Specific Answers	Solve for target time, limit, or amount.
55–60 m	Audit & Header	Check units, box final answers, sign IDs.

- [!TIP] **Team Division Strategy:**
- **Student A (Formulator / Lead):** Sets up the differential equation, verifies signs (e.g., $k < 0$ in cooling, minus sign on outflow), and tracks boundary conditions.
 - **Student B (Algebraist / Integrator):** Executes the calculus (integrating factor, definite/indefinite integrals, fractions).

- **Student C (Validator / Auditor):** Recalculates constants on a calculator, performs dimensional unit analysis, and checks limits as $t \rightarrow \infty$.

2. Core Mathematical Archetypes Summary

LINEAR MODELING MASTER FORMULARY (ZILL 2.7)		
Archetype	Differential Equation	Master Integrated Solution
Growth / Decay	$dP/dt = k P$	$P(t) = P_0 * \exp(k*t)$
Newton's Cooling	$dT/dt = k (T - T_m)$	$T(t) = T_m + (T_0 - T_m) * \exp(k*t)$
Tank (Equal Flow)	$dA/dt + (Q/V_0) A = c_{in} * Q$	$A(t) = c_{in} * V_0 + (A_0 - c_{in}*V_0) * \exp(-(Q/V_0)*t)$
Tank (Unequal Flow)	$dA/dt + [Q_{out}/(V_0+dQ*t)] A = c_{in} * Q_{in} \quad (dQ = Q_{in} - Q_{out})$	Integrate using $\mu(t) = [V_0 + dQ*t]^{(Q_{out}/dQ)}$
LR Series Circuit	$L (di/dt) + R i = E_0$	$i(t) = (E_0/R) * [1 - \exp(-(R/L)*t)] \quad (\text{if } i(0) = 0)$
RC Series Circuit	$R (dq/dt) + (1/C) q = E_0$	$q(t) = C*E_0 * [1 - \exp(-t/(R*C))] \quad (\text{if } q(0) = 0)$

3. Deep Archetype Analysis & Step-by-Step Derivations

Archetype 1: Exponential Growth, Decay & Half-Life

- **Governing Equation:** $\frac{dP}{dt} = kP \implies P(t) = P_0 e^{kt}$.
- **Solving for Rate Constant k :** Given $P(t_1) = P_1$:

$$k = \frac{1}{t_1} \ln \left(\frac{P_1}{P_0} \right)$$

- **Doubling Time:** $P(t) = 2P_0 \implies t_{\text{double}} = \frac{\ln 2}{k}$.
- **Half-Life ($t_{1/2}$):** $P(t) = \frac{1}{2}P_0 \implies t_{1/2} = \frac{\ln(1/2)}{k} = -\frac{\ln 2}{k} \implies k = -\frac{\ln 2}{t_{1/2}}$.

Archetype 2: Newton's Law of Cooling / Warming

- **Governing Equation:** $\frac{dT}{dt} = k(T - T_m)$.
- **General Solution:** $T(t) = T_m + (T_0 - T_m)e^{kt}$.
- **Crucial Sign Rules:**
 - In heat loss ($T > T_m$), $\frac{dT}{dt} < 0 \implies k < 0$.
 - In heat absorption ($T < T_m$), $(T - T_m) < 0$ and $\frac{dT}{dt} > 0 \implies k < 0$.
 - Therefore, **k is always strictly negative.**
- **Variable Ambient Temperature $T_m(t)$:** If the ambient temperature changes with time (e.g., $T_m(t) = T_0 + A \sin(\omega t)$ or linear warming), the ODE becomes non-autonomous:

$$\frac{dT}{dt} - kT = -kT_m(t)$$

Solve using integrating factor $\mu(t) = e^{-kt}$.

Archetype 3: Tank Mixture Problems (The #1 Tutorial Favorite)

- **General Mass Balance:**

$$\frac{dA}{dt} = R_{in} - R_{out} = c_{in}Q_{in} - c_{out}(t)Q_{out}$$

- **Instantaneous Tank Volume:**

$$V(t) = V_0 + (Q_{in} - Q_{out})t = V_0 + \Delta Q \cdot t$$

- **Instantaneous Outflow Concentration:**

$$c_{out}(t) = \frac{A(t)}{V(t)} = \frac{A(t)}{V_0 + \Delta Q \cdot t}$$

Sub-Case 3A: Equal Flow Rates ($Q_{in} = Q_{out} = Q \implies \Delta Q = 0$)

- Volume remains constant: $V(t) = V_0$.
- ODE: $\frac{dA}{dt} + \frac{Q}{V_0}A = c_{in}Q$.
- Integrating factor: $\mu(t) = e^{\frac{Q}{V_0}t}$.
- Analytic Solution:

$$A(t) = c_{in}V_0 + (A_0 - c_{in}V_0)e^{-\frac{Q}{V_0}t}$$

- Steady-state limit: $\lim_{t \rightarrow \infty} A(t) = c_{in}V_0$.

Sub-Case 3B: Unequal Flow Rates ($Q_{in} \neq Q_{out}$)

- Volume is time-dependent: $V(t) = V_0 + (Q_{in} - Q_{out})t$.
- ODE: $\frac{dA}{dt} + \frac{Q_{out}}{V_0 + \Delta Qt}A = c_{in}Q_{in}$.
- Integrating factor:

$$P(t) = \frac{Q_{out}}{V_0 + \Delta Qt} \implies \int P(t)dt = \frac{Q_{out}}{\Delta Q} \ln(V_0 + \Delta Qt)$$

$$\mu(t) = e^{\ln \left[(V_0 + \Delta Qt)^{\frac{Q_{out}}{\Delta Q}} \right]} = (V_0 + \Delta Qt)^{\frac{Q_{out}}{\Delta Q}}$$

- **Critical Boundary Check:**

- If $Q_{in} > Q_{out}$, tank fills up! Valid only until overflow: $t_{\text{overflow}} = \frac{V_{\text{capacity}} - V_0}{Q_{in} - Q_{out}}$.
- If $Q_{in} < Q_{out}$, tank empties! Valid only until dry: $t_{\text{empty}} = \frac{V_0}{Q_{out} - Q_{in}}$.

Sub-Case 3C: Pure Water Flush ($c_{in} = 0$)

- Input rate is zero: $R_{in} = 0$.
- $\frac{dA}{dt} = -\frac{Q}{V_0}A \implies A(t) = A_0 e^{-\frac{Q}{V_0}t}$ (pure exponential decay).

Archetype 4: Series Electric Circuits (LR and RC)

- **Kirchhoff's Second Law:** $\sum V_{\text{drops}} = E(t)$.

LR Series Circuit:

$$L \frac{di}{dt} + Ri = E(t) \implies \frac{di}{dt} + \frac{R}{L}i = \frac{E(t)}{L}$$

- Integrating factor: $\mu(t) = e^{\frac{R}{L}t}$.
- For constant DC voltage $E(t) = E_0$ and $i(0) = 0$:

$$i(t) = \frac{E_0}{R} \left(1 - e^{-\frac{R}{L}t}\right)$$

- Steady-state current: $i_{ss} = \frac{E_0}{R}$. Transient current: $i_{tr}(t) = -\frac{E_0}{R}e^{-\frac{R}{L}t}$.
- Time constant: $\tau = \frac{L}{R}$ [seconds].

RC Series Circuit:

$$R \frac{dq}{dt} + \frac{1}{C}q = E(t) \quad \text{with } i(t) = \frac{dq}{dt}$$

- Integrating factor: $\mu(t) = e^{\frac{t}{RC}}$.
- For constant DC voltage E_0 and $q(0) = 0$:

$$q(t) = CE_0 \left(1 - e^{-\frac{t}{RC}}\right)$$

$$i(t) = \frac{dq}{dt} = \frac{E_0}{R} e^{-\frac{t}{RC}}$$

- Capacitive time constant: $\tau = RC$ [seconds].

4. Master Fully Solved Practice Problems (Tutorial & Exam Prep)

Problem 1 (Bacterial Culture with Variable Benchmark)

Statement: A bacterial culture grows at a rate proportional to the population present. Initially, there are 500 bacteria. After 3 hours, the population reaches 1,500.

- (a) Find an explicit formula for population $P(t)$ at any time t (in hours).
- (b) What will the population be after 8 hours?
- (c) When will the population reach 10,000?

Step-by-Step Solution:

- **Part (a):**

$$\frac{dP}{dt} = kP \implies P(t) = P_0 e^{kt}$$

Given $P_0 = 500$:

$$P(t) = 500e^{kt}$$

At $t = 3$, $P(3) = 1500$:

$$1500 = 500e^{3k} \implies e^{3k} = 3 \implies 3k = \ln 3 \implies k = \frac{\ln 3}{3} \approx 0.366204 \text{ h}^{-1}$$

Exact expression:

$$P(t) = 500e^{\left(\frac{\ln 3}{3}\right)t} = 500 \left(3^{t/3}\right)$$

- **Part (b):** At $t = 8$ hours:

$$P(8) = 500 \left(3^{8/3}\right) = 500 \times 3^{2.6667} \approx 500 \times 18.72075 \approx 9,360 \text{ bacteria}$$

- **Part (c):** Set $P(t) = 10,000$:

$$10000 = 500 \left(3^{t/3}\right) \implies 3^{t/3} = \frac{10000}{500} = 20$$

Take natural logarithm on both sides:

$$\frac{t}{3} \ln 3 = \ln 20 \implies t = 3 \frac{\ln 20}{\ln 3} = 3 \times \frac{2.99573}{1.09861} \approx 8.18 \text{ hours (8 h 11 min)}$$

Problem 2 (Radioactive Half-Life & Forensic Isotope Decay)

Statement: A radioactive isotope decays at a rate proportional to the mass remaining. A sample initially has a mass of 120 mg. After 24 hours, its mass has decayed to 90 mg.

- (a) Determine the decay constant k and the half-life $t_{1/2}$.
- (b) How much mass remains after 4 days?
- (c) How long will it take for 95 of the original mass to decay?

Step-by-Step Solution:

- **Part (a):**

$$\frac{dA}{dt} = kA \implies A(t) = A_0 e^{kt} = 120e^{kt}$$

At $t = 24$ hours:

$$90 = 120e^{24k} \implies e^{24k} = \frac{90}{120} = 0.75 \implies 24k = \ln(0.75)$$

$$k = \frac{\ln(0.75)}{24} = \frac{-0.287682}{24} \approx -0.0119868 \text{ h}^{-1}$$

Half-life $t_{1/2}$:

$$t_{1/2} = \frac{-\ln 2}{k} = \frac{-0.693147}{-0.0119868} \approx 57.83 \text{ hours (approx 2.41 days)}$$

- **Part (b):** 4 days = 96 hours:

$$A(96) = 120e^{(-0.0119868)(96)} = 120e^{-1.15073} = 120 \times 0.316406 \approx 37.97 \text{ mg}$$

(Notice: $96 \text{ h} = 4 \times 24 \text{ h}$, so $A = 120 \times (0.75)^4 = 120 \times 0.316406 = 37.97 \text{ mg}$).

- **Part (c):** When 95 has decayed, 5 remains:

$$A(t) = 0.05 \times 120 = 6 \text{ mg}$$

$$120e^{kt} = 6 \implies e^{kt} = 0.05 \implies kt = \ln(0.05)$$

$$t = \frac{\ln(0.05)}{-0.0119868} = \frac{-2.99573}{-0.0119868} \approx 249.92 \text{ hours (approx 10.41 days)}$$

Problem 3 (Forensic Cooling & Time of Death)

Statement: A body is found in a room kept at a constant temperature of 68°F . At 9:00 AM, the body temperature is measured to be 85°F . At 11:00 AM, the temperature drops to 79°F . Assuming normal human body temperature at time of death was 98.6°F , estimate the time of death.

Step-by-Step Solution:

- **Step 1: Coordinate Frame:** Let $t = 0$ correspond to 9:00 AM.
 - $T_m = 68^\circ\text{F}$
 - $T(0) = 85^\circ\text{F}$
 - $T_0 - T_m = 85 - 68 = 17^\circ\text{F}$
 - General formula: $T(t) = 68 + 17e^{kt}$
- **Step 2: Determine k using 11:00 AM measurement ($t = 2$ hours):**

$$T(2) = 79 \implies 68 + 17e^{2k} = 79 \implies 17e^{2k} = 11$$

$$e^{2k} = \frac{11}{17} \implies 2k = \ln\left(\frac{11}{17}\right) = -0.435318$$

$$k = \frac{-0.435318}{2} \approx -0.217659 \text{ h}^{-1}$$

- **Step 3: Solve for Time of Death t_d when $T(t_d) = 98.6^\circ\text{F}$:**

$$68 + 17e^{kt_d} = 98.6 \implies 17e^{kt_d} = 30.6 \implies e^{kt_d} = \frac{30.6}{17} = 1.8$$

$$kt_d = \ln(1.8) \implies t_d = \frac{\ln(1.8)}{-0.217659} = \frac{0.587787}{-0.217659} \approx -2.7005 \text{ hours}$$

- **Step 4: Clock Conversion:** -2.7005 hours represents 2 hours and $(0.7005 \times 60) \text{ min} = 2 \text{ hours } 42 \text{ minutes before 9:00 AM}$:

$$9:00 \text{ AM} - 2 \text{ h } 42 \text{ min} = \mathbf{6 : 18 \text{ AM}}$$

Problem 4 (Brine Tank with Pure Water Washout)

Statement: A tank holds 200 gal of brine containing 40 lb of dissolved salt. Pure water is pumped into the tank at 4 gal/min, and the thoroughly mixed solution is pumped out at the same rate.

- (a) Find the amount of salt $A(t)$ in the tank at any time t .
- (b) What is the salt concentration in the tank after 30 minutes?

- (c) At what time will the salt content be reduced to 5 lb?

Step-by-Step Solution:

- **Part (a):**

- $V_0 = 200$ gal, $Q_{in} = Q_{out} = 4$ gal/min $\implies V(t) = 200$ gal.
- Inflow is pure water: $c_{in} = 0 \implies R_{in} = 0$.
- Outflow rate: $R_{out} = c_{out}Q_{out} = \left(\frac{A}{200}\right) \times 4 = \frac{A}{50}$ lb/min.

$$\frac{dA}{dt} = -\frac{A}{50} \implies \frac{dA}{A} = -\frac{1}{50}dt$$

$$A(t) = A_0 e^{-t/50} = 40 e^{-t/50} \text{ lb}$$

- **Part (b):** At $t = 30$ min:

$$A(30) = 40 e^{-30/50} = 40 e^{-0.6} = 40 \times 0.54881 = 21.95 \text{ lb}$$

Concentration:

$$c(30) = \frac{A(30)}{V_0} = \frac{21.95 \text{ lb}}{200 \text{ gal}} \approx \mathbf{0.1098 \text{ lb/gal}}$$

- **Part (c):** Set $A(t) = 5$ lb:

$$40 e^{-t/50} = 5 \implies e^{-t/50} = \frac{5}{40} = \frac{1}{8}$$

$$-\frac{t}{50} = \ln\left(\frac{1}{8}\right) = -\ln 8 \implies t = 50 \ln 8 = 50(2.07944) \approx \mathbf{103.97 \text{ minutes}}$$

Problem 5 (Unequal Flow Rates — Accumulating Tank with Overflow)

Statement: A 500-gallon tank initially contains 200 gallons of pure water. A brine solution containing 3 lb/gal of salt is pumped in at 4 gal/min. The well-stirred mixture is pumped out at 2 gal/min.

- (a) Find the instantaneous volume $V(t)$ and the time when the tank begins to overflow.
- (b) Find the amount of salt $A(t)$ in the tank prior to overflow.
- (c) Determine the exact amount of salt in the tank at the instant of overflow.

Step-by-Step Solution:

- **Part (a): Volume & Overflow Time:**

$$V(t) = V_0 + (Q_{in} - Q_{out})t = 200 + (4 - 2)t = 200 + 2t \text{ gallons}$$

Tank capacity is 500 gallons. Overflow occurs when $V(t) = 500$:

$$200 + 2t = 500 \implies 2t = 300 \implies t_{\text{overflow}} = \mathbf{150 \text{ minutes}}$$

- **Part (b): Differential Equation Formulation:**

- Inflow rate: $R_{in} = c_{in}Q_{in} = (3 \text{ lb/gal}) \times (4 \text{ gal/min}) = 12 \text{ lb/min}$.
- Outflow concentration: $c_{out}(t) = \frac{A(t)}{200+2t}$.

- Outflow rate: $R_{out} = c_{out}Q_{out} = \frac{A(t)}{200+2t} \times 2 = \frac{2}{200+2t}A = \frac{1}{100+t}A$.

$$\frac{dA}{dt} + \frac{1}{100+t}A = 12, \quad A(0) = 0$$

- **Integrating Factor Calculation:**

$$\mu(t) = e^{\int \frac{1}{100+t} dt} = e^{\ln(100+t)} = 100 + t$$

- **Integration:**

$$\frac{d}{dt} [A(t)(100 + t)] = 12(100 + t)$$

$$A(t)(100 + t) = 12 \int (100 + t) dt = 12 \left[100t + \frac{t^2}{2} \right] + C = 1200t + 6t^2 + C$$

$$A(t) = \frac{1200t + 6t^2 + C}{100 + t}$$

- **Initial Condition** $A(0) = 0$:

$$0 = \frac{0 + C}{100} \implies C = 0$$

$$A(t) = \frac{1200t + 6t^2}{100 + t} = \frac{6t(200 + t)}{100 + t} \text{ lb, for } 0 \leq t \leq 150 \text{ min}$$

- **Part (c): Salt at Overflow** ($t = 150$ min):

$$A(150) = \frac{1200(150) + 6(150)^2}{100 + 150} = \frac{180000 + 6(22500)}{250} = \frac{180000 + 135000}{250}$$

$$A(150) = \frac{315000}{250} = \mathbf{1,260 \text{ lb}}$$

- Concentration at overflow: $c(150) = \frac{1260 \text{ lb}}{500 \text{ gal}} = \mathbf{2.52 \text{ lb/gal}}$.

Problem 6 (LR Series Circuit with Periodic AC Voltage)

Statement: An LR-series circuit contains an inductor $L = 1 \text{ H}$ and resistor $R = 10 \Omega$. An alternating voltage $E(t) = 100 \sin(10t) \text{ V}$ is applied. The initial current is $i(0) = 0$.

- Find the current $i(t)$ and identify the steady-state and transient components.

Step-by-Step Solution:

- **Step 1: Standard Linear ODE:**

$$1 \frac{di}{dt} + 10i = 100 \sin(10t) \implies \frac{di}{dt} + 10i = 100 \sin(10t)$$

- **Step 2: Integrating Factor:**

$$\mu(t) = e^{\int 10 dt} = e^{10t}$$

$$\frac{d}{dt} [ie^{10t}] = 100e^{10t} \sin(10t)$$

- **Step 3: Evaluate Integral** $\int e^{at} \sin(bt) dt$: Using the standard formula $\int e^{at} \sin(bt) dt = \frac{e^{at}}{a^2 + b^2} [a \sin(bt) - b \cos(bt)]$: Here $a = 10, b = 10 \implies a^2 + b^2 = 100 + 100 = 200$:

$$\int 100e^{10t} \sin(10t) dt = 100 \times \frac{e^{10t}}{200} [10 \sin(10t) - 10 \cos(10t)] = 5e^{10t} [\sin(10t) - \cos(10t)]$$

- **Step 4: Form General Solution:**

$$i(t)e^{10t} = 5e^{10t} [\sin(10t) - \cos(10t)] + C$$

$$i(t) = 5 \sin(10t) - 5 \cos(10t) + Ce^{-10t}$$

- **Step 5: Apply Initial Condition** $i(0) = 0$:

$$0 = 5(0) - 5(1) + C \implies C = 5$$

$$i(t) = \underbrace{5 \sin(10t) - 5 \cos(10t)}_{\text{Steady-State } i_{ss}(t)} + \underbrace{5e^{-10t}}_{\text{Transient } i_{tr}(t)} \text{ Amperes}$$

Problem 7 (RC Series Circuit Charging Dynamics)

Statement: A constant voltage $E_0 = 100 \text{ V}$ is applied to a series circuit with resistance $R = 200 \Omega$ and capacitance $C = 10^{-4} \text{ F}$ ($100 \mu\text{F}$). If $q(0) = 0$:

- (a) Find charge $q(t)$ and current $i(t)$.
- (b) Find the maximum charge on the capacitor.
- (c) Calculate the time required for the capacitor to reach 90 of its maximum charge.

Step-by-Step Solution:

- **Part (a):**

$$R \frac{dq}{dt} + \frac{1}{C} q = E_0 \implies 200 \frac{dq}{dt} + \frac{1}{10^{-4}} q = 100 \implies 200 \frac{dq}{dt} + 10000 q = 100$$

Divide by 200:

$$\frac{dq}{dt} + 50q = 0.5$$

Integrating factor: $\mu(t) = e^{50t}$.

$$\frac{d}{dt} [qe^{50t}] = 0.5e^{50t} \implies q(t)e^{50t} = \frac{0.5}{50} e^{50t} + C_1 = 0.01e^{50t} + C_1$$

$$q(t) = 0.01 + C_1 e^{-50t}$$

Using $q(0) = 0 \implies C_1 = -0.01$:

$$q(t) = 0.01 (1 - e^{-50t}) \text{ Coulombs}$$

Current $i(t) = \frac{dq}{dt}$:

$$i(t) = \frac{d}{dt} [0.01 - 0.01e^{-50t}] = (-0.01)(-50)e^{-50t} = \mathbf{0.5e^{-50t}} \text{ Amperes}$$

- **Part (b): Maximum Charge:**

$$\lim_{t \rightarrow \infty} q(t) = q_{\max} = CE_0 = (10^{-4} \text{ F}) \times (100 \text{ V}) = \mathbf{0.01} \text{ Coulombs (10 mC)}$$

- **Part (c): 90% Charge Horizon:**

$$q(t) = 0.90 \times 0.01 = 0.009 \text{ C}$$

$$0.01(1 - e^{-50t}) = 0.009 \implies 1 - e^{-50t} = 0.9 \implies e^{-50t} = 0.1$$

$$-50t = \ln(0.1) = -\ln(10) \implies t = \frac{\ln 10}{50} = \frac{2.302585}{50} \approx \mathbf{0.04605} \text{ seconds (46.1 ms)}$$

Problem 8 (Two-Stage Environmental Reservoir Clean-up)

Statement: A chemical spill introduces 500 kg of a toxic industrial solvent into a 10,000 m³ settling pond. Remediation begins immediately by pumping clean water in at 50 m³/hour and discharging treated water at the same rate. Safe environmental discharge mandates that solvent concentration must drop below 0.005 kg/m³.

- Determine how many hours and days the remediation pumps must operate continuously to achieve the safe threshold.

Step-by-Step Solution:

- **Step 1: Model Setup:**

- Volume: $V_0 = 10,000 \text{ m}^3$, $Q = 50 \text{ m}^3/\text{h}$ (constant volume).
- Initial solvent: $A_0 = 500 \text{ kg}$.
- Initial concentration: $c_0 = \frac{500}{10000} = 0.05 \text{ kg/m}^3$.
- Clean water inflow: $c_{in} = 0 \implies R_{in} = 0$.
- Outflow rate: $R_{out} = \left(\frac{A}{10000}\right) \times 50 = \frac{A}{200} \text{ kg/h}$.

- **Step 2: Solve ODE:**

$$\frac{dA}{dt} = -\frac{A}{200} \implies A(t) = 500e^{-t/200} \text{ kg}$$

Instantaneous concentration:

$$c(t) = \frac{A(t)}{10000} = \frac{500e^{-t/200}}{10000} = 0.05e^{-t/200} \text{ kg/m}^3$$

- **Step 3: Solve for Target Threshold $c(t) = 0.005 \text{ kg/m}^3$:**

$$0.05e^{-t/200} = 0.005 \implies e^{-t/200} = \frac{0.005}{0.05} = 0.1$$

$$-\frac{t}{200} = \ln(0.1) = -\ln 10 \implies t = 200 \ln 10$$

$$t = 200 \times 2.302585 = \mathbf{460.52} \text{ hours}$$

- **Step 4: Day Conversion:**

$$\text{Days} = \frac{460.52}{24} \approx \mathbf{19.19} \text{ days (approx 19 days and 5 hours)}$$

5. Team Project 1 Pre-Submission Checklist

Before submitting your team's solution paper to the tutorial TA, verify each item:

- [] **All Team Members Listed:** Full Legal Names and 8-Digit Student ID Numbers clearly written at top right.
 - [] **Model Stated Explicitly:** Did you write the differential equation with correct physical signs (e.g., $k < 0$, $-R_{out}$)?
 - [] **Integrating Factor Complete:** Did you show the calculation of $\mu(t) = e^{\int P(t)dt}$?
 - [] **Units on Every Constant:**
 - Rate constant k has units [1/time] (h^{-1} , min^{-1} , s^{-1}).
 - Mass $A(t)$ has units [lb] or [kg].
 - Current $i(t)$ has units [A], charge $q(t)$ has units [C].
 - [] **Sanity Check on Limits:** Does your solution make physical sense as $t \rightarrow \infty$?
 - Mixture salt amount should approach $c_{in}V$.
 - Object temperature should approach T_m .
 - Current in LR circuit should approach E_0/R .
 - [] **Boxed Final Answers:** Clearly box your final numerical answers and time horizons.
-